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# Cheap Measures, Scarce Validity

Partial Identification of Admissible Measures  
for Latent Economic Constructs

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*Position Paper v2* · *An alternative positioning of the SCA 2.0 research program*

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**Relationship to v1.** This is a deliberate *re-positioning*, not a revision. Version 1 (SCA2\_PositionPaper.tex) leads with the synthetic-agent pipeline and treats measurement as the validation layer. This version inverts the hierarchy: the measurement problem is the paper, and AI-generated measures—including our own fine-tuned adapters—are candidate measures that must earn admission on the same terms as survey items. All numbers are identical to v1 and unchanged; the argument, the ordering, and the claimed contribution differ. Current empirical results are **proof-of-concept**. Code: [github.com/EconLLM-Lab/SCA2\\_PofW](https://github.com/EconLLM-Lab/SCA2_PofW); instrument: [github.com/Bonorinoa/cvprofiles](https://github.com/Bonorinoa/cvprofiles).

## Statement of Significance

Economics runs on quantities nobody can observe: trust, patience, risk tolerance. Researchers pick a survey item, defend it in a footnote, and proceed. That practice was tolerable when measures were scarce and expensive. Artificial intelligence has made them abundant and nearly free—any model can now be asked to score any construct—so the binding constraint has moved from *building* measures to *knowing which ones are admissible*. We formalize the classical construct-validity network as a partial-identification problem: a researcher states the theoretical relations any valid measure must satisfy, and an open-source instrument returns the set of measures that survive plus the range of any downstream conclusion consistent with all of them. Applied to two major preference surveys, only 5 of 16 candidate measures survive, and the canonical trust item is not among them.

## Abstract

The measurement of latent constructs in economics has an unacknowledged degree of freedom: the choice of proxy. We show this choice is consequential and propose a discipline for it. Formalizing the Cronbach–Meehl nomological network [5] as a system of moment inequalities, we recast construct validity as *partial identification* [11] over a finite menu of candidate measures: given researcher-authored restrictions with literature-anchored thresholds, the admissible set  $\mathcal{M}^*$  collects measures satisfying every restriction, and the construct-identified range  $[\hat{L}, \hat{U}]$  reports how much a downstream estimate moves across admissible measures. The framework is implemented in the open-source package *cvprofiles*. Applied to the Global Preferences Survey (GPS) and World Values Survey (WVS), only 5 of 16 candidate survey facets are admissible representations of the six GPS preference dimensions; three dimensions admit *no* measure. The canonical

single-item trust question (Q57) is rejected, and it disagrees with the surviving composite about *who* trusts—the cross-country correlation of gender gaps is 0.005 for Q57 versus 0.568 for the composite—while the trust–development correlation ranges over  $[0.371, 0.624]$  across admissible measures. We then argue the framework’s natural scope includes AI-generated measures, and demonstrate with fine-tuned language-model adapters that emit choice distributions over survey items: a Mexico-trained adapter improves on its base model against Mexican survey responses (better on 97.1% of 35 items) while a US-trained adapter degrades, illustrating that admissibility, not provenance, is the right screen for machine-generated measures.

**Keywords:** construct validity; partial identification; measurement; latent constructs; economic preferences; AI-generated data

## 1. The Measurement Bottleneck

Trust, patience, risk attitudes, and social preferences are not observable. They are recovered through instruments—survey items, incentivized tasks, experimental games—each an imperfect projection of the attribute onto a number. The standard practice is to select one instrument, justify it briefly, and treat the resulting variable as the construct. The justification is rarely tested, and the sensitivity of conclusions to the choice is rarely reported.

This was defensible under scarcity. Fielding a new measure cost money and time, so the menu of candidates was short and the choice was often forced. That constraint has now dissolved. Language models can generate a plausible score for any construct, on any unit, at negligible cost [10, 7]; the menu is effectively unbounded. Dell and Rambachan [7] put the resulting problem precisely: when AI makes candidate measures cheap, the bottleneck becomes *choosing among many plausible representations*, which may yield different estimates of the target parameter. Abundance has converted a scarcity problem into a selection problem, and economics has no standard procedure for the latter.

**The argument in one line.** Validity is not a property a measure has or lacks; it is a *restriction* a measure either satisfies or violates. Once stated as restrictions, validity becomes identification—and the honest output is a set, not a point.

Psychometrics solved the conceptual half of this problem seventy years ago and economics has largely ignored the solution. Cronbach and Meehl [5] argued that when no criterion exists, a construct is defined by its *nomological network*: the system of theoretical relations linking it to other constructs and observables. A measure is valid to the extent that it reproduces the network’s implied relations. Campbell and Fiske [4] operationalized this as convergent and discriminant validity; Borsboom et al. [1] added the essential caveat that a network establishes *consistency*, not existence or causation.

Econometrics solved the other half. Leamer [9] asked how much conclusions move across specifications; Manski [11, 12] formalized the discipline of reporting the *set* of parameter values consistent with the data and the stated assumptions. The measurement-error literature supplies machinery for mismeasured and multi-measure systems [2, 16].

Our contribution is to observe that these are the same problem and to make the connection operational. A nomological network *is* a set of assumptions; its restrictions *are* moment inequalities; the set of measures satisfying them *is* an identified set; and the image of a downstream estimand over that set is the identified set for the conclusion. This paper states the correspondence formally (Section 2), implements it in an open-source instrument, applies it to two major preference surveys (Section 3), and extends it to AI-generated measures (Section 4).

**What this reframing buys.** Three things a single-proxy practice cannot deliver. *First*, a verdict: measures that fail the network are named and their violations quantified, rather than being absorbed into a robustness appendix. *Second*, a range: the analyst learns how much of a conclusion is attributable to measurement choice. *Third*, a screen that does not care where a measure came from—the same restrictions adjudicate a survey item and a neural network, which is what makes the framework durable as AI-generated measures proliferate.

## 2. Construct Validity as Partial Identification

Let  $X$  denote the underlying reality—respondents, countries, their behaviors—and let a *measurement function*  $m_j(X)$  be any disciplined projection of  $X$  into a score. The researcher supplies three objects.

1. A **menu**  $\mathcal{M} = \{m_j(\cdot)\}_{j \in \mathcal{J}}$  of candidate measures, normalized so measures and downstream estimands are comparable across candidates.
2. A **nomological network**: a set  $\mathcal{R}$  of moment restrictions, each stating a behavior required of any valid measure,

$$r \in \mathcal{R} : \quad \mathbb{E}[g_r(m(X), V; \theta_r)] \geq 0, \quad (1)$$

where  $V = (Z, W, G, Q)$  collects auxiliary observables— $Z$  related quantities (convergent evidence),  $W$  distinct quantities (discriminant),  $G$  groups that should differ (known-groups),  $Q$  conditioning covariates—and  $\theta_r$  is a threshold *anchored in published evidence* rather than data-mined.

3. A **tolerance**  $\delta \geq 0$  (default 0).

The *slack* of measure  $m$  on restriction  $r$  is the sample analogue of (1),  $s_r(m; \theta_r) = \widehat{\mathbb{E}}[g_r(m(X), V; \theta_r)]$ , positive when satisfied and negative by the magnitude of violation. The **admissible set** is

$$\widehat{\mathcal{M}}_{\mathcal{R}, \theta}^* = \{m \in \mathcal{M} : s_r(m; \theta_r) \geq -\delta \ \forall r \in \mathcal{R}\}. \quad (2)$$

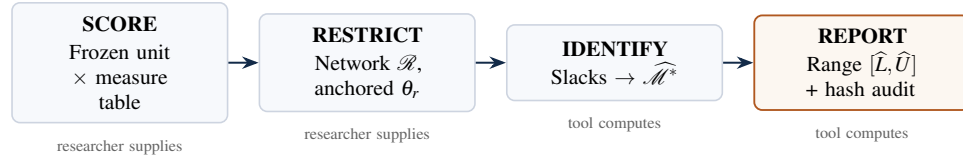
For a downstream target functional  $\beta(\cdot)$ —a correlation, regression coefficient, or treatment effect defined using the measure—the **construct-identified range** is

$$[\widehat{L}, \widehat{U}] = \left[ \inf_{m \in \widehat{\mathcal{M}}^*} \beta(m), \sup_{m \in \widehat{\mathcal{M}}^*} \beta(m) \right]. \quad (3)$$

**Interpretation.** Equation (2) is a sample object: a screening rule applied to observed data under researcher-specified restrictions, encoding *conceptual* uncertainty about representation. It is not a population identified set unless slacks are replaced by their population counterparts, and neither object is exhaustive unless the menu and the restrictions are themselves defended as sufficiently rich. This is a real limitation and we state it plainly: the framework disciplines the choice *within* a stated menu; it cannot certify that the menu contains the right measure.

**Inference and reproducibility.** Sampling uncertainty is handled by bootstrap over units—the menu is fixed, not resampled—reporting percentile bands on range endpoints over non-empty replicates, with degenerate replicates counted and reported. Threshold sensitivity is a multiplicative  $\lambda$ -grid diagnostic ( $\lambda = 1$  is the headline). Reported numbers require frozen scores, a pinned network and target, a fixed seed, and a package version, with a deterministic run id derived from their hash. Empty admissible sets and wide ranges are valid outputs; thresholds are never loosened to avoid them.

**Implementation.** The framework is implemented in `cvprofiles` [6], a score-agnostic, model-free engine whose state machine is exactly SCORE  $\rightarrow$  RESTRICT  $\rightarrow$  IDENTIFY  $\rightarrow$  REPORT (Figure 1). It ingests a frozen



**Figure 1: The instrument separates judgment from computation.** The researcher owns the menu and the network—the scientific content. The tool owns slacks, the admissible set, and the range—the bookkeeping. Restrictions are stated before scores are seen, and empty sets are reported rather than repaired.

unit-by-measure score table, binds a researcher-authored network and target, and emits JSON/HTML audit artifacts under run-id and hash discipline. It contains no model calls; the network is always researcher-authored and the tool only evaluates.

### 3. Application I: Survey Measures of Economic Preferences

The GPS [8] supplies incentivized, behaviorally validated measures of patience, risk taking, positive and negative reciprocity, altruism, and trust across 76 countries. The WVS supplies candidate survey measures of nominally the same constructs. This is an unusually clean setting: two independent survey traditions, one with behavioral validation, targeting a shared set of latent attributes.

**Setup.** The construct is country-level generalized trust; the unit is a country ( $n = 42$  in the  $\text{WVS} \cap \text{GPS}$  overlap). The menu is four WVS Wave 7 trust facets, each scored from respondent-level items under a frozen protocol and aggregated to country means. The network is a single convergent restriction,  $\text{Corr}(m, \text{gps\_trust}) \geq 0.30$ , with the threshold anchored in published survey-behavioral trust correlations ( $r \approx 0.29$ , OECD [14]; 0.28–0.35 country-level in [8]). Run id 483fc619641476a9.

**Table 1:** Construct-validity profile, trust. Unit = country ( $n = 42$ );  $\delta = 0$ ; slack =  $r - 0.30$ .

| Measure                   | Items           | $\text{Corr}(m, \text{GPS})$ | Slack  | Verdict                 |
|---------------------------|-----------------|------------------------------|--------|-------------------------|
| trust-general (Q57)       | Q57             | 0.278                        | −0.022 | <b>rejected</b>         |
| trust-in-group            | Q58, Q60        | 0.349                        | +0.049 | admissible              |
| trust-out-group           | Q59, Q61–63     | 0.303                        | +0.003 | admissible (knife-edge) |
| trust-institution         | Q64, Q69–71     | 0.332                        | +0.032 | admissible              |
| <b>Survivor composite</b> | in + out + inst | 0.410                        | —      | (derived reference)     |

The single most-used trust item in empirical economics—Q57, “most people can be trusted”—is *rejected* at the literature-anchored threshold, while three multi-item facets survive. Their mean, the survivor composite, correlates 0.41 with GPS trust. A  $\lambda$ -grid confirms this is not knife-edge overall: at  $\lambda = 0.5$  all four facets pass; at  $\lambda = 1.0$ , three; at  $\lambda = 1.5$  and  $\lambda = 2.0$  the set is empty—an honest report that at that standard the network and data admit no measure at all.

**Across all six dimensions.** Table 2 applies the same machinery to the full menu. Only 5 of 16 candidate facets are admissible. Three dimensions—risk taking, altruism, positive reciprocity—have *empty* admissible sets at the headline threshold: their mapped WVS items are not admissible representations of the incentivized constructs under this network. Patience admits a single measure, and a supplementary age-shape diagnostic flags even that one (financial satisfaction, Q50) as U-shaped where GPS patience is hump-shaped—admissible at country level, structurally wrong at individual level.

**Table 2:** Outcome typology across the six GPS dimensions ( $\theta = 0.30$ ,  $\delta = 0$ ).

| Dimension    | Menu      | $ \widehat{\mathcal{N}}^* $ | Admissible measures              |
|--------------|-----------|-----------------------------|----------------------------------|
| trust        | 4         | 3                           | in-group, out-group, institution |
| patience     | 3         | 1                           | financial (Q50)                  |
| negrecip     | 3         | 1                           | justifiability (Q177, Q179)      |
| risktaking   | 2         | 0                           | <i>empty</i>                     |
| altruism     | 1         | 0                           | <i>empty</i>                     |
| posrecip     | 3         | 0                           | <i>empty</i>                     |
| <b>total</b> | <b>16</b> | <b>5</b>                    |                                  |

**Headline result.** Most survey measures of economic preferences are not admissible representations of the incentivized constructs they are used to proxy. The framework says so in an auditable, threshold-transparent way—and reports the three empty sets as findings rather than suppressing them.

### 3.1 Representation choice is consequential, not cosmetic

A natural objection is that admissibility is academic: different trust measures presumably rank countries similarly, so conclusions are safe. They do not.

**Same average, different people.** Q57 and the survivor composite agree on *average* trust across countries but disagree about *who* has it. At country level they correlate 0.28 and 0.41 with GPS trust. But the cross-country correlation of WVS-vs-GPS *gender gaps* is 0.005 for Q57—effectively no structural alignment—versus 0.568 for the composite (country-cluster bootstrap,  $P(\text{composite} > \text{Q57}) = 0.999$ ). At the level of sex-by-age cells within countries, Q57’s pooled within-country correlation with GPS trust is +0.087 with a confidence interval crossing zero, versus +0.334 for the composite (CI [0.234, 0.415]). Any model that interacts trust with demographic composition will therefore obtain GPS-inconsistent trust effects when fed the canonical item.

**The range on a real conclusion.** On public data ( $n = 35$  countries; WVS  $\times$  GPS  $\times$  World Bank) with a richer three-restriction network— $\text{corr}(m, \text{gps\_trust}) \geq 0.3$ ,  $\text{corr}(m, \text{rule\_of\_law}) \geq 0.3$ ,  $\text{corr}(m, \text{gini}) \leq -0.1$ —and target  $\beta(m) = \text{Corr}(m, \log \text{GDP per capita})$  deliberately excluded from the network, two measures survive and both designed-invalid controls are rejected (false-admission rate 0). The construct-identified range of the trust–development correlation is

$$[\widehat{L}, \widehat{U}] = [0.371, 0.624], \quad (4)$$

with a units bootstrap band of [0.174, 0.752] and 17.5% empty replicates reported. A quarter of a correlation point in one of development economics’ most-cited relationships is attributable to measurement choice alone—a quantity the single-proxy convention cannot express.

## 4. Application II: AI-Generated Measures Enter the Menu

Nothing in Section 2 restricts  $m_j$  to survey items. A measurement function is any disciplined projection of  $X$  into a score, and a language model that emits a probability distribution over survey response options is exactly that. This is the framework’s most consequential property: as AI-generated measures proliferate, they can be screened by the same restrictions that screen survey items, with no special pleading for their provenance.

**Constructing an AI measure.** We build candidate measures  $m_{AI}$  by encoding a measured cultural state into a model’s weights. A six-dimensional GPS state vector  $z_c$ —standardized deviations from the global mean—generates culture-conditioned synthetic preference pairs, where the sign of  $z_c$  on the target dimension selects which of two fixed behavioral options is labeled preferred. A student model (Llama-3.1-8B-Instruct) is fine-tuned on these pairs by DPO under QLoRA. Two design choices matter for measurement. First, the country is *never named*, at training or inference: cultural information enters only through which response was preferred, so the measure cannot lean on country-name stereotypes. Second, the resulting adapter is scored by option likelihood— $\hat{p}_k \propto \exp(\log p(y_k | x))$ —so it is used as a choice-probability simulator, not a text generator. Under the Bradley–Terry reading of DPO [15, 3, 13], implied reward margins are model-implied utility differences, identified up to scale and location; we do not claim they are separately identified human utilities.

**Evidence.** Against Mexican WVS responses on  $N = 35$  unseen Wave 7 items, the Mexico-trained adapter beats the cross-country adapter on 97.1% of items (34/35; mean  $\Delta\text{TVD} = 0.1495$ , 95% CI [0.0964, 0.2152]) and improves on the base model (0.4882 vs. 0.5211). Because prompts never name a country, this difference is carried entirely by the weights. The US-trained adapter behaves oppositely: it is worse than its own base model (0.5258 vs. 0.3958) and worse than the Mexico adapter on US data, with the mechanism—collision with a strong American pretraining prior versus overfitting to scenario format—not identified by this design.

**Table 3:** Out-of-sample evaluation against human survey distributions,  $N = 35$  unseen WVS Wave 7 items, unconditioned prompts. Lower TVD is better.

| Target     | Measure             | TVD ↓         | JSD ↓         | vs. cross            |
|------------|---------------------|---------------|---------------|----------------------|
| Mexico wvs | base model          | 0.5211        | 0.2165        | —                    |
| Mexico wvs | <b>MEX adapter</b>  | <b>0.4882</b> | <b>0.2010</b> | <b>97.1% (34/35)</b> |
| Mexico wvs | USA adapter (cross) | 0.6377        | 0.2965        | —                    |
| USA wvs    | <b>base model</b>   | <b>0.3958</b> | <b>0.1439</b> | —                    |
| USA wvs    | MEX adapter (cross) | 0.3888        | 0.1405        | —                    |
| USA wvs    | USA adapter         | 0.5258        | 0.2246        | 11.4% (4/35)         |

**The reframing.** Under a pipeline-first reading, the US adapter is an embarrassment—a component that failed. Under a measurement-first reading it is the instrument working correctly: a candidate measure was constructed, screened against an external human criterion, and *should not be admitted*. The framework’s value is precisely that it issues this verdict without regard to the measure’s provenance or to the effort that produced it.

**An honest boundary.** We have not yet run a formal `cvprofiles` network over the adapter menu; the comparison above uses out-of-sample distributional fit against human responses rather than a stated set of moment restrictions. Constructing that network—with convergent, discriminant, and known-groups restrictions on AI measures—is the immediate next step, and it is what would convert the informal verdict in the callout into an admissible set. We flag this gap rather than papering over it.

**Two standing caveats.** Option-likelihood scoring exhibits severe under-dispersion (bias  $-1.15$  to  $-1.63$ ): softmax concentrates mass on modal choices, so claims about *absolute shares* are unsupported while claims about *direction and rank* are. And self-recovery on held-out synthetic pairs (Mexico 78.5%, USA 70.0% preference accuracy at  $n = 200$  per cell, versus 52.0% and 45.5% cross) is in-sample evidence only, since



training labels and evaluation items share generative ancestry. The external criterion in Table 3 is what answers the correlated-errors objection to validating AI measures against AI measures [7].

## 5. Discussion

**What the framework establishes.** That validity claims can be stated as restrictions, evaluated reproducibly, and summarized as a set; that on two major preference surveys most candidate measures fail; and that the failures change downstream conclusions by economically meaningful margins. Following Borsboom et al. [1], passing a network establishes consistency with that network—not the construct’s existence, nor the direction of causation between attribute and score.

**What it does not.** It does not certify that a menu is complete or that a network is correct. Both are researcher inputs, and a poorly chosen network will admit poor measures with full auditability. The framework relocates judgment to a place where it is visible and criticizable; it does not eliminate it. Nor does the current evidence support strong claims about the AI measures: the adapter panel is two countries, the mechanism behind the US failure is unresolved, and the measurement layer’s 42-country coverage does not extend to the adapter layer.

**Reliable uses today.** Quantifying measurement fragility in an existing literature; pre-field stress-testing of draft survey items against a target benchmark; power analysis across  $[\hat{L}, \hat{U}]$  rather than a single assumed effect size; and directional comparative statics from AI measures that have passed an external check. **Unreliable:** absolute headcount or polling claims, free-form persona dialogue, and granular per-cell elasticities.

**Agenda.** Four steps, in order of leverage. (i) Multi-restriction networks per dimension, adding discriminant and known-groups restrictions to the current convergent-only profiles, with a downstream  $\beta$  so every dimension reports a range. (ii) A formal admissibility network for AI measures, which would let adapters, base models, and steered policies [17, 18] be screened on common terms; because steering intensity is continuous, the menu becomes a continuum and admissibility a function of steering strength. (iii) A third country whose GPS profile differs from Mexico’s, to test the prior-collision account of the US failure. (iv) A public construct registry, where research groups contribute score matrices and networks under anchored-threshold and frozen-run discipline, so that measurement validity accumulates as a community resource rather than a per-paper judgment. The endpoint we find most interesting: if restrictions can screen measures, they can also *train* them—fine-tuning gated on satisfying a stated nomological network would yield models aligned to theory rather than to annotator preference.

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